

기초전기실험

15주차

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15 주차 실험 목표

▣ AC Ch. 14 : Parallel resonant circuits

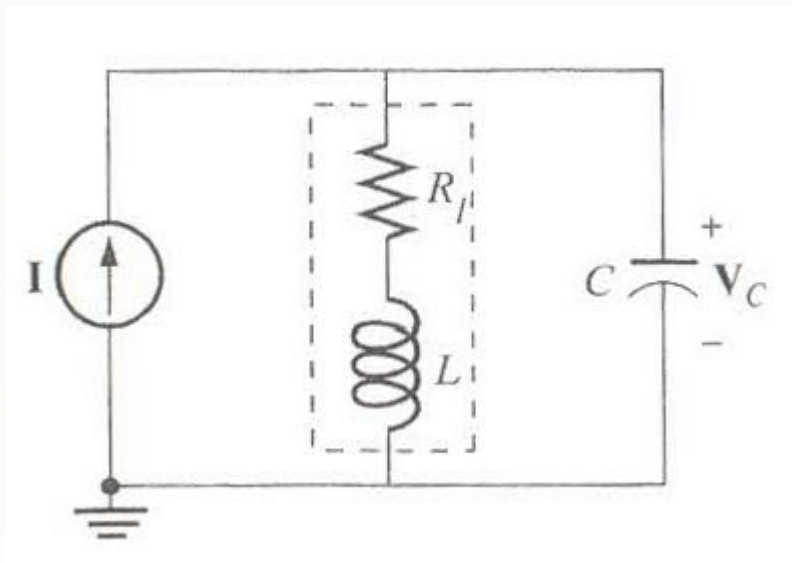
- ▣ 병렬로 R-L-C가 연결되어 있을 때 주파수 변화에 따른 전압 및 전류를 이론적으로 계산해 보고, 실험을 통해 확인한다.
- ▣ Resonant frequency를 이론적으로 계산하고, 실험을 통해 확인한다.
- ▣ 주파수 변화에 따른 input impedance를 측정한다.
- ▣ Quality factor와 bandwidth의 관계를 확인한다.

▣ AC Ch. 15 : Passive filters

- ▣ R-C를 이용한 high-pass filter를 제작하고 실험을 통해 확인한다.
- ▣ R-L-C를 이용한 band-pass filter를 제작하고 실험을 통해 확인한다.

PARALLEL RESONANT CIRCUITS

▣ Parallel Resonant Circuits



Admittance

$$\begin{aligned} Y &= \frac{1}{R_l + jX_L} + \frac{1}{-jX_C} \\ &= \frac{R_l - jX_L}{R_l^2 + X_L^2} + \frac{j}{X_C} \\ &= \frac{R_l}{R_l^2 + X_L^2} + j \left(\frac{1}{X_C} - \frac{X_L}{R_l^2 + X_L^2} \right) \end{aligned}$$

At resonance

At resonance, the impedance is resistive, so the susceptance must be zero.

We obtain

$$\frac{1}{X_C} = \frac{X_L}{R_l^2 + X_L^2}$$

PARALLEL RESONANT CIRCUITS

▣ Parallel Resonant Circuits

Admittance

$$Y = \frac{R_l}{R_l^2 + X_L^2} + j \left(\frac{1}{X_c} - \frac{X_L}{R_l^2 + X_L^2} \right)$$

The impedance at resonance is the inverse of the admittance at resonance:

$$Z_{Tp} = \frac{R_l^2 + X_L^2}{R_l} = \frac{X_L X_C}{R_l} = \frac{\omega_p L \left(\frac{1}{\omega_p C} \right)}{R_l} = \frac{L}{R_l C}$$

$$\frac{1}{X_c} = \frac{X_L}{R_l^2 + X_L^2}$$

PARALLEL RESONANT CIRCUITS

▣ Parallel Resonant Circuits

The resonant frequency

$$\frac{1}{X_c} = \frac{X_L}{R_l^2 + X_L^2}$$

$$\omega_p C = \frac{\omega_p L}{R_l^2 + (\omega_p L)^2}$$

$$\omega_p^2 = \frac{1}{LC} - \frac{R_l^2}{L^2} = \frac{1}{LC} \left(1 - \frac{R_l^2 C}{L} \right)$$

$$f_p = \frac{1}{2\pi\sqrt{LC}} \sqrt{1 - \frac{R_l^2 C}{L}}$$

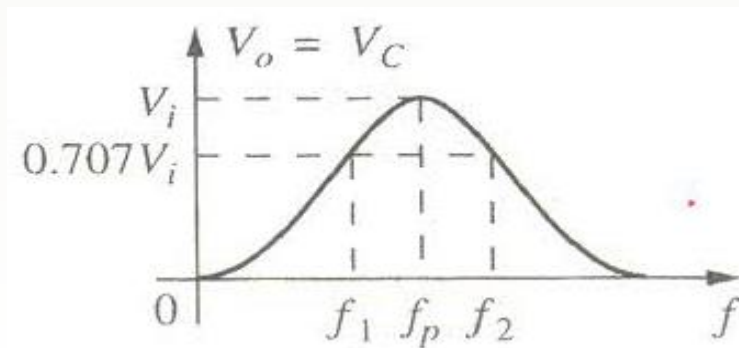
PARALLEL RESONANT CIRCUITS

▣ Parallel Resonant Circuits

Q-factor(전류 확대도, 선택도)

$$Q = \frac{I_C}{I} = \frac{V/X_C}{V/Z_{Tp}} = \frac{Z_{Tp}}{X_C} = \frac{\frac{R_l^2 + X_L^2}{R_l}}{X_C} = \frac{\frac{L}{R_l C}}{1/\omega_p C} = \frac{\omega_p L}{R_l} = \frac{X_L}{R_l}$$

Frequency Response



$$\text{Bandwidth : } f_2 - f_1 = \frac{f_p}{Q}$$

PARALLEL RESONANT CIRCUITS

▣ Parallel Resonant Circuits

$$\frac{1}{X_C} = \frac{X_L}{R_l^2 + X_L^2} \longrightarrow X_C = \frac{R_l^2 + X_L^2}{X_L} = \frac{X_L}{\frac{X_L^2}{R_l^2}} + X_L = \frac{X_L}{Q_{coil}^2} + X_L$$

If Q_{coil} is large enough $\longrightarrow X_C \cong X_L$

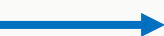
$$f_p = \frac{1}{2\pi\sqrt{LC}} \sqrt{1 - \frac{R_l^2 C}{L}} \longrightarrow \frac{1}{\frac{\omega L}{R_l^2 \omega C}} = \frac{1}{\frac{X_L X_C}{R_l^2}} \cong \frac{1}{\frac{X_L^2}{R_l^2}} = \frac{1}{Q_{coil}^2} \cong 0$$

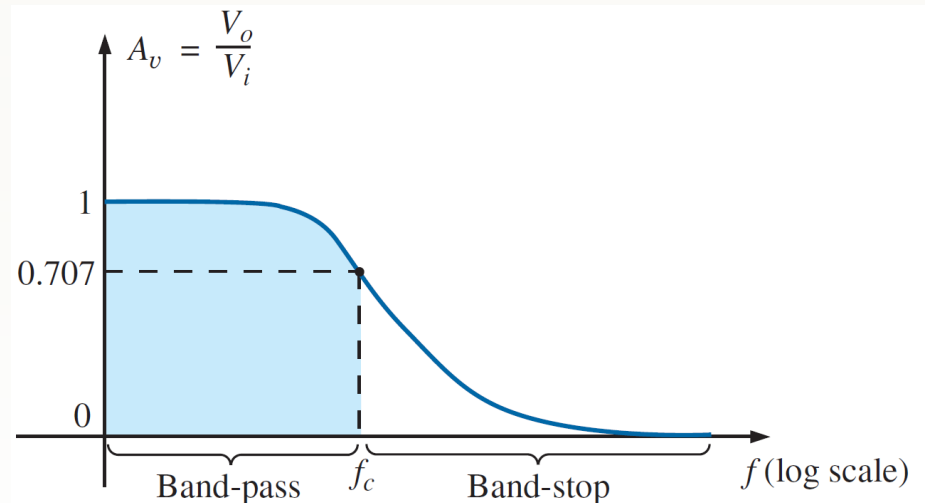
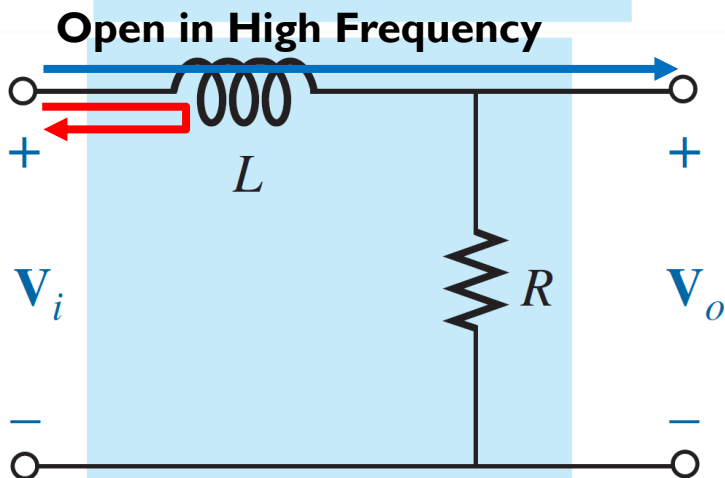
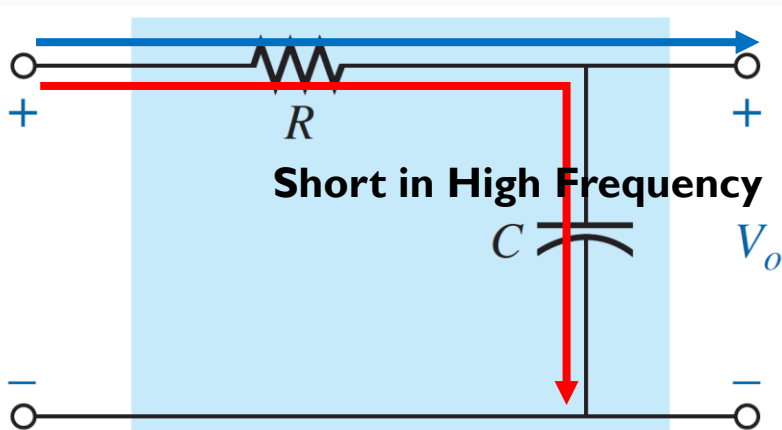
$$\therefore f_p = \frac{1}{2\pi\sqrt{LC}} \quad \text{if } Q_{coil} = \frac{X_L}{R_l} \geq 10$$

PASSIVE FILTER

Low-Pass Filter(LPF, 저역통과필터)

$$\times X_L = 2\pi fL \quad X_C = \frac{1}{2\pi fC}$$

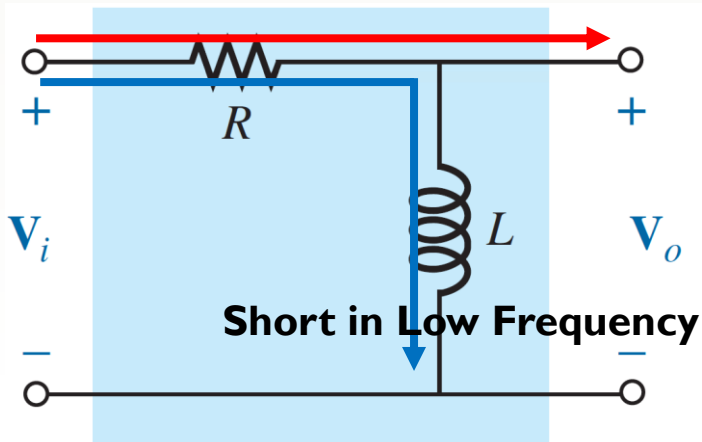
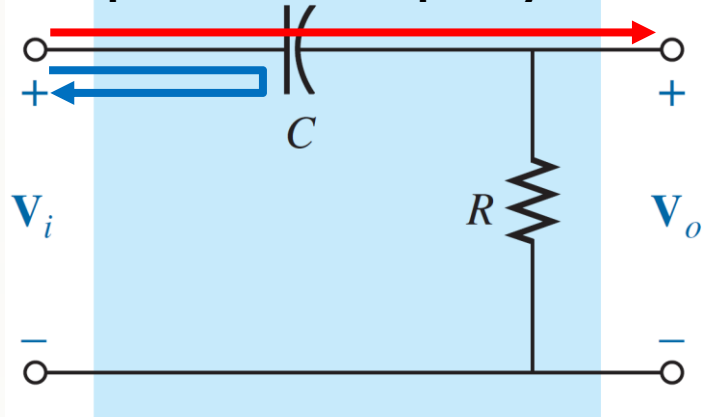
High Frequency : 
Low Frequency : 



PASSIVE FILTER

High-Pass Filter(HPF, 고역통과필터)

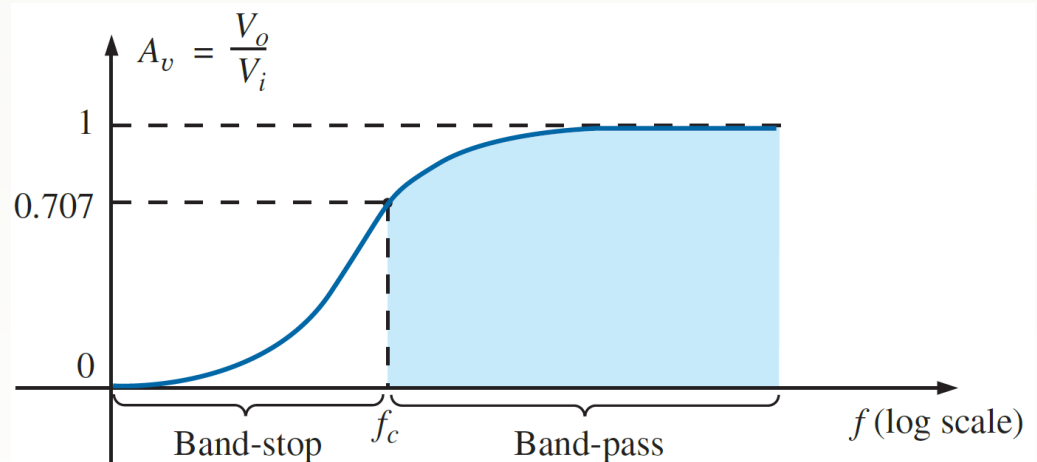
Open in Low Frequency



$$\ast X_L = 2\pi fL \quad X_C = \frac{1}{2\pi fC}$$

High Frequency :

Low Frequency :

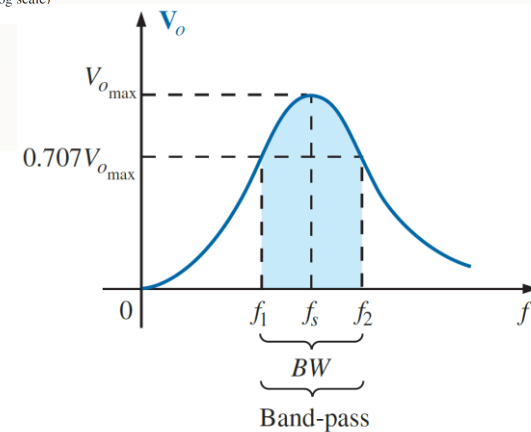
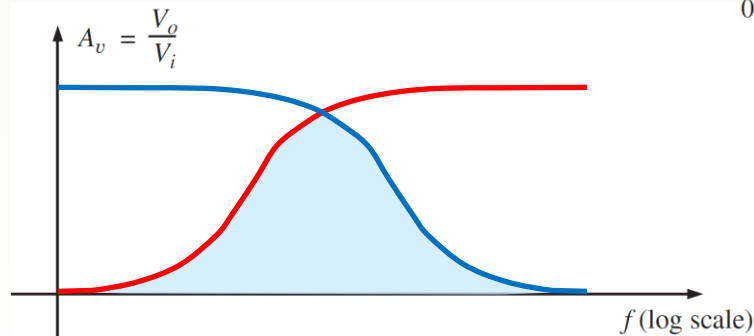
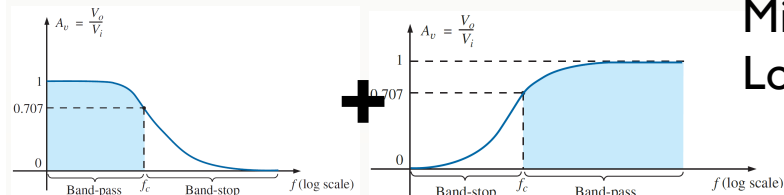
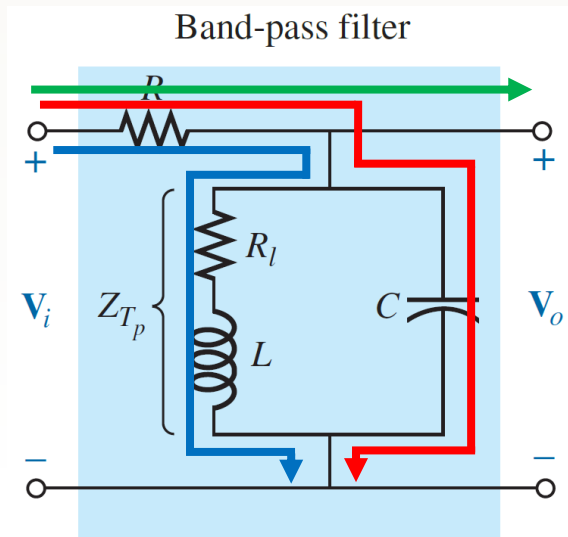
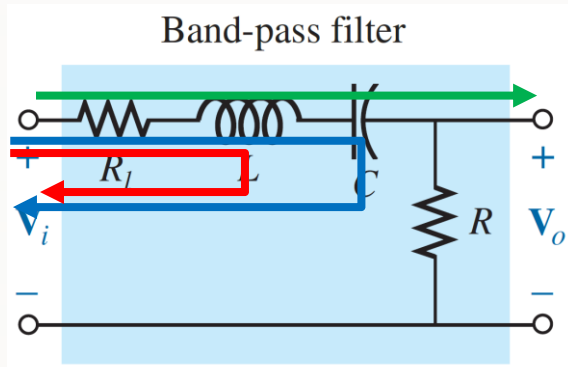


PASSIVE FILTER

Band-Pass Filter(BPF, 대역통과필터)

$$\times X_L = 2\pi fL \quad X_C = \frac{1}{2\pi fC}$$

High Frequency : →
 Middle Range : →
 Low Frequency : →



PASSIVE FILTER

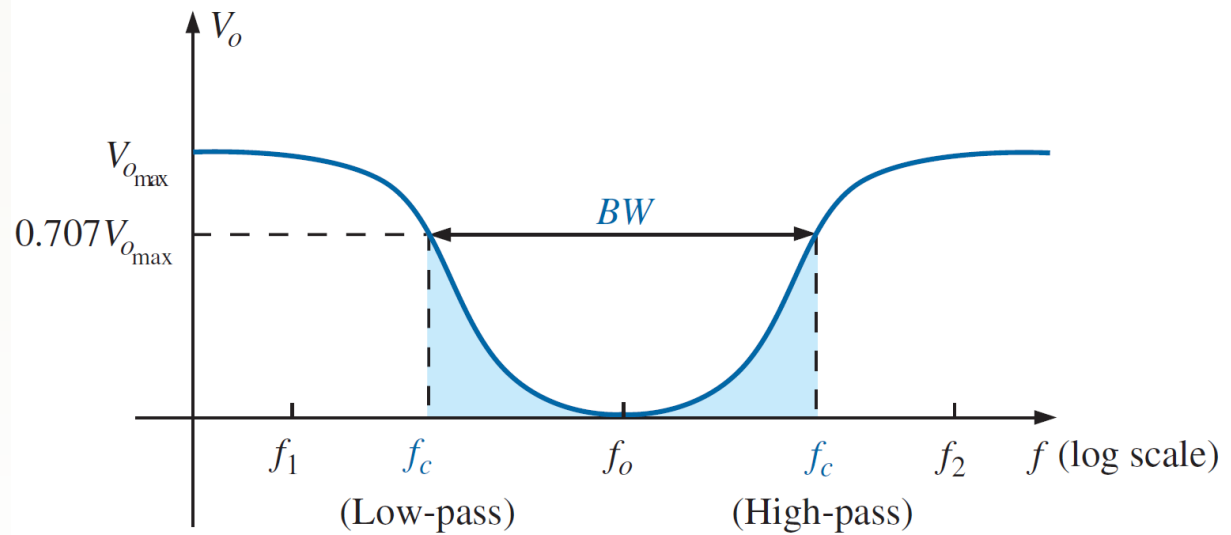
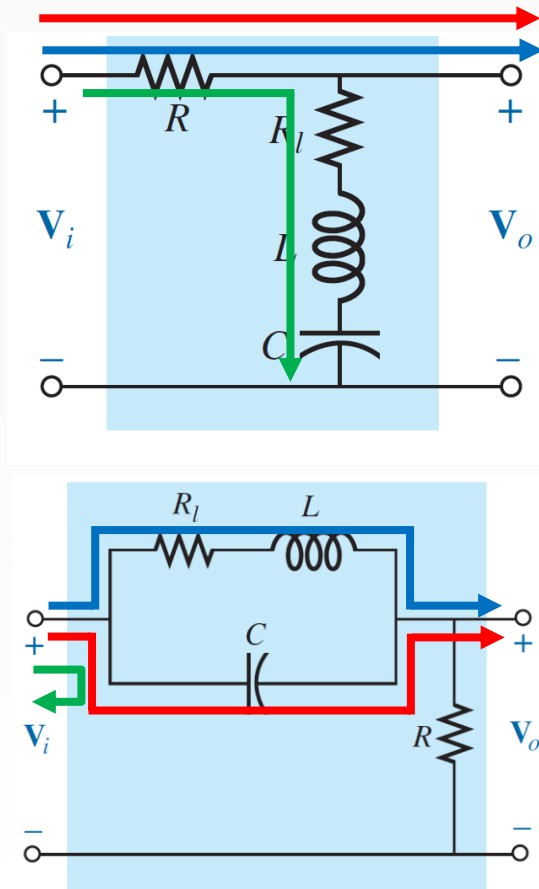
■ Band-stop Filter(BSF, 대역저지필터)

$$\ast X_L = 2\pi fL \quad X_C = \frac{1}{2\pi fC}$$

High Frequency : 

Middle Range : 

Low Frequency : 



※ Band-Reject Filter(BRF)로도 불림

15 주차 실험

▣ AC Ch. 14 : Parallel resonant circuits

- ▣ part1($100\text{k}\Omega \times 1$, $4.7\Omega \times 1$, $10\text{mH} \times 1$, $0.1\mu\text{F} \times 1$)
 - ▣ FIG 14.2 구성 후 $E_s = 5V_{p-p}$ (500~10,000Hz) 인가
 - ▣ 주파수별 R_s, C 의 전압 측정 후 Table 14.2완성

▣ AC Ch. 15 : Passive filters

- ▣ part1($1\text{k}\Omega \times 1$, $0.1\mu\text{F} \times 1$)
 - ▣ FIG 15.4 구성 후 $E_s = 5V_{p-p}$ (0.1~100kHz) 인가
 - ▣ 주파수별 R 의 전압 V_o , 위상차 측정 후 Table 15.2, 15.3완성
- ▣ part2($10\text{mH} \times 1$, $0.1\mu\text{F} \times 1$, $100\Omega \times 1$)
 - ▣ FIG 15.5 구성 후 $E_s = 5V_{p-p}$ (0.1~100kHz) 인가
 - ▣ 주파수별 R 의 전압 V_o 측정 후 Table 15.5완성
- ▣ Part3($1\text{mH} \times 1$, $0.2\mu\text{F} \times 1$, $100\Omega \times 1$)
 - ▣ FIG 15.6 구성 후 $E_s = 5V_{p-p}$ (0.1~100kHz) 인가
 - ▣ 주파수별 R 의 전압 V_o 측정 후 Table 15.7완성